Frontend Lab Assignment - 3

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1. Create a BMI calculator using JS. Take height and weight as input, calculate BMI and display the category (underweight / normal / overweight)

Code (bmi.html):

```
&lt;!DOCTYPE html&gt;
&lt;html&gt;
&lt;head&gt;
&lt;meta charset="UTF-8"&gt;
&lt;title&gt;BMI Calculator&lt;/title&gt;
&lt;style&gt;
  body { font-family: Arial, sans-serif; background: #f2f2f2; text-align: center; }
  .box { background: white; width: 320px; margin: 60px auto; padding: 25px; border: 1px solid #999; }
  input { width: 90%; padding: 8px; margin: 8px 0; }
  button { padding: 8px 20px; cursor: pointer; }
  #result { margin-top: 15px; font-size: 18px; }
&lt;/style&gt;
&lt;/head&gt;
&lt;body&gt;
  &lt;div class="box"&gt;
    &lt;h2&gt;BMI Calculator&lt;/h2&gt;
    &lt;input type="number" id="height" placeholder="Height in cm"&gt;&lt;br&gt;
    &lt;input type="number" id="weight" placeholder="Weight in kg"&gt;&lt;br&gt;
    &lt;button onclick="calculateBMI()"&gt;Calculate&lt;/button&gt;
    &lt;div id="result"&gt;&lt;/div&gt;
  &lt;/div&gt;

&lt;script&gt;
function calculateBMI() {
  var h = document.getElementById("height").value;
  var w = document.getElementById("weight").value;
  var result = document.getElementById("result");

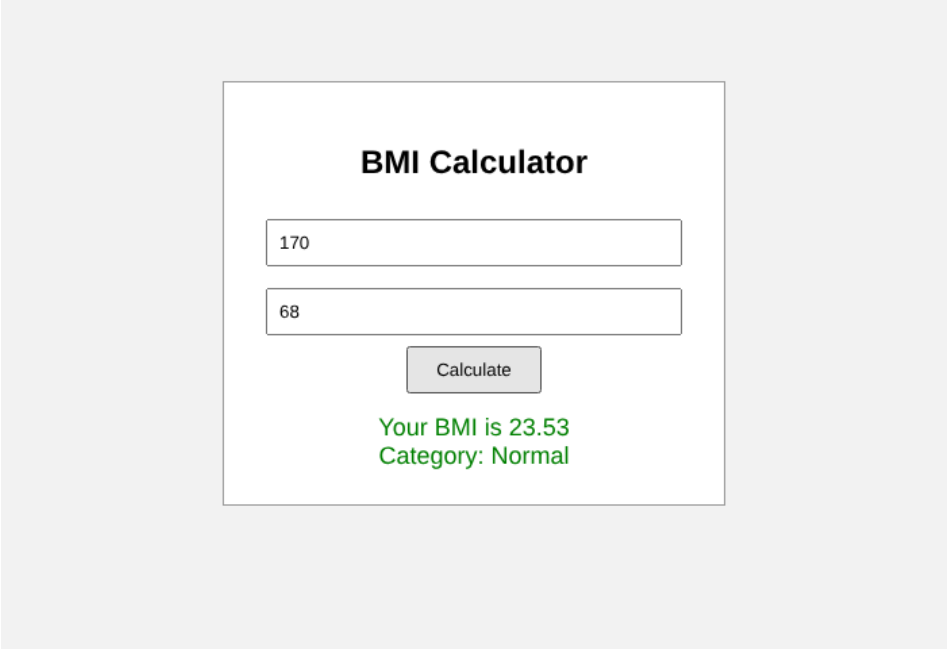
  if (h &lt;= 0 || w &lt;= 0 || h === "" || w === "") {
    result.innerHTML = "Please enter valid height and weight";
    result.style.color = "red";
    return;
  }

  var heightInMeter = h / 100;
  var bmi = w / (heightInMeter * heightInMeter);
  var category = "";
  var color = "";

  if (bmi &lt; 18.5) {
    category = "Underweight";
    color = "blue";
  } else if (bmi &lt; 25) {
    category = "Normal";
    color = "green";
  } else {
    category = "Overweight";
    color = "red";
  }

  result.innerHTML = "Your BMI is " + bmi.toFixed(2) + "&lt;br&gt;Category: " + category;
  result.style.color = color;
}
&lt;/script&gt;
&lt;/body&gt;
&lt;/html&gt;
```

Output:


BMI Calculator
170
68
Calculate
Your BMI is 23.53
Category: Normal

2. Make a signup page and apply form validations

Code (signup.html):

```
&lt;!DOCTYPE html&gt;
&lt;html&gt;
&lt;head&gt;
&lt;meta charset="UTF-8"&gt;
&lt;title&gt;Signup Page&lt;/title&gt;
&lt;style&gt;
  body { font-family: Arial, sans-serif; background: #f2f2f2; }
  .box { background: white; width: 340px; margin: 40px auto; padding: 25px; border: 1px solid #999; }
  label { display: block; margin-top: 10px; }
  input { width: 95%; padding: 7px; margin-top: 4px; }
  .error { color: red; font-size: 13px; height: 14px; }
  button { margin-top: 15px; padding: 8px 20px; cursor: pointer; }
  #success { color: green; margin-top: 10px; }
&lt;/style&gt;
&lt;/head&gt;
&lt;body&gt;
  &lt;div class="box"&gt;
    &lt;h2&gt;Signup&lt;/h2&gt;
    &lt;form id="signupForm"&gt;
      &lt;label&gt;Full Name&lt;/label&gt;
      &lt;input type="text" id="name"&gt;
      &lt;div class="error" id="nameError"&gt;&lt;/div&gt;

      &lt;label&gt;Email&lt;/label&gt;
      &lt;input type="text" id="email"&gt;
      &lt;div class="error" id="emailError"&gt;&lt;/div&gt;

      &lt;label&gt;Phone Number&lt;/label&gt;
      &lt;input type="text" id="phone"&gt;
      &lt;div class="error" id="phoneError"&gt;&lt;/div&gt;

      &lt;label&gt;Password&lt;/label&gt;
      &lt;input type="password" id="password"&gt;
      &lt;div class="error" id="passwordError"&gt;&lt;/div&gt;

      &lt;label&gt;Confirm Password&lt;/label&gt;
      &lt;input type="password" id="confirm"&gt;
      &lt;div class="error" id="confirmError"&gt;&lt;/div&gt;

      &lt;button type="submit"&gt;Sign Up&lt;/button&gt;
    &lt;/form&gt;
    &lt;div id="success"&gt;&lt;/div&gt;
  &lt;/div&gt;

&lt;script&gt;
document.getElementById("signupForm").addEventListener("submit", function (e) {
  e.preventDefault();

  var name = document.getElementById("name").value.trim();
  var email = document.getElementById("email").value.trim();
  var phone = document.getElementById("phone").value.trim();
  var password = document.getElementById("password").value;
  var confirm = document.getElementById("confirm").value;
  var valid = true;

  // clear old messages
  var errors = document.getElementsByClassName("error");
  for (var i = 0; i &lt; errors.length; i++) {
    errors[i].innerHTML = "";
  }
  document.getElementById("success").innerHTML = "";

  if (name.length &lt; 3) {
    document.getElementById("nameError").innerHTML = "Name must be at least 3 characters";
    valid = false;
  }

  var emailPattern = /^[^\s@]+@[^\s@]+\.[^\s@]+$/;
  if (!emailPattern.test(email)) {
    document.getElementById("emailError").innerHTML = "Enter a valid email";
    valid = false;
  }

  var phonePattern = /^[0-9]{10}$/;
  if (!phonePattern.test(phone)) {
```

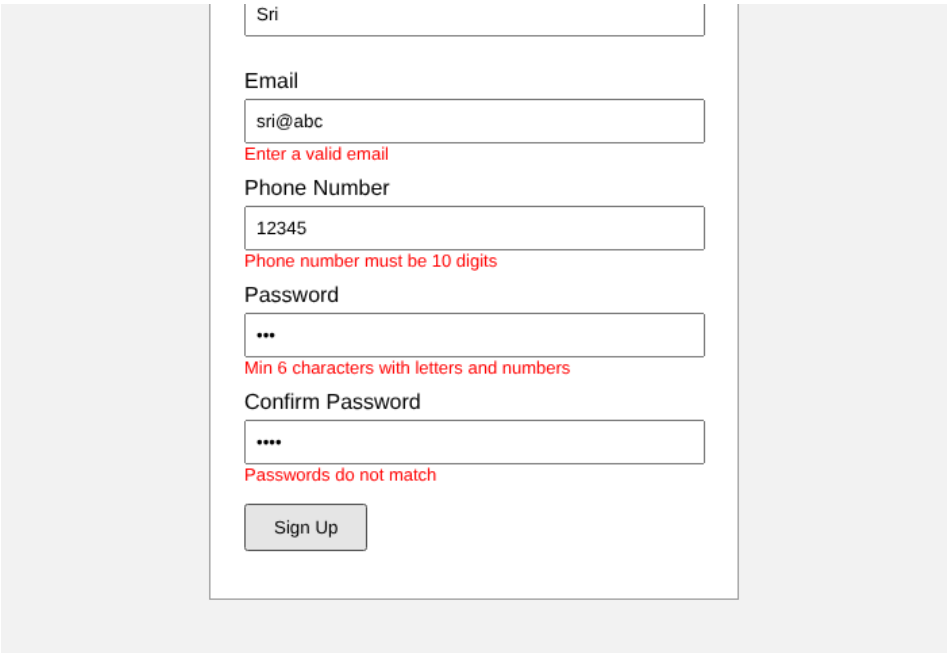
```
    document.getElementById("phoneError").innerHTML = "Phone number must be 10 digits";
    valid = false;
  }

  var passPattern = /^(?=.*[A-Za-z])(?=.*[0-9]).{6,}$/;
  if (!passPattern.test(password)) {
    document.getElementById("passwordError").innerHTML = "Min 6 characters with letters and numbers";
    valid = false;
  }

  if (confirm !== password || confirm === "") {
    document.getElementById("confirmError").innerHTML = "Passwords do not match";
    valid = false;
  }

  if (valid) {
    document.getElementById("success").innerHTML = "Signup successful!";
    document.getElementById("signupForm").reset();
  }
});
&lt;/script&gt;
&lt;/body&gt;
&lt;/html&gt;
```

Output:



3. Create a page in which user can change color theme.

Code (theme.html):

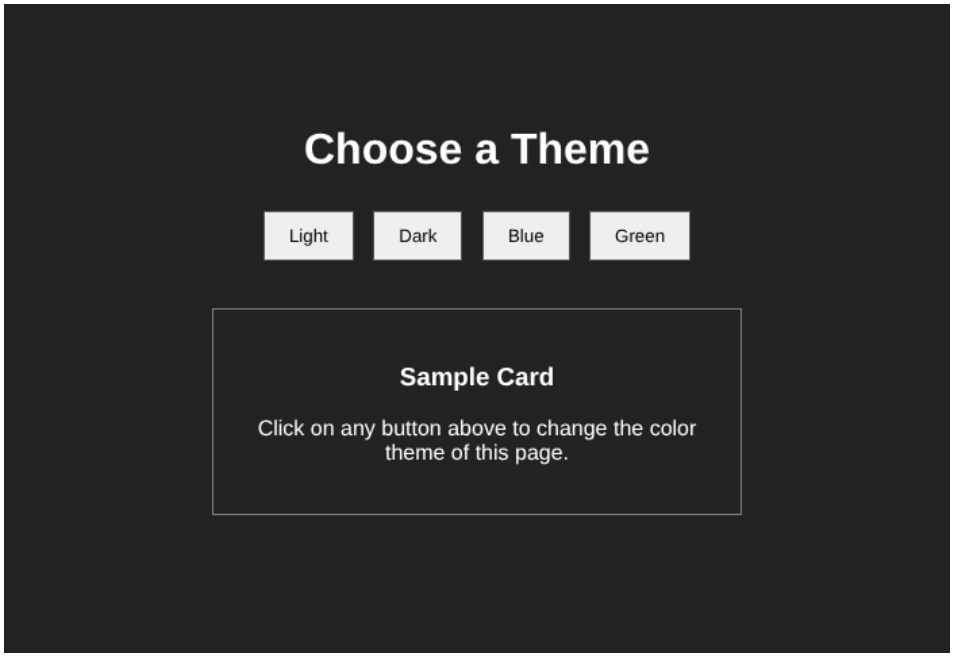
```
&lt;!DOCTYPE html&gt;
&lt;html&gt;
&lt;head&gt;
&lt;meta charset="UTF-8"&gt;
&lt;title&gt;Change Color Theme&lt;/title&gt;
&lt;style&gt;
  body { font-family: Arial, sans-serif; text-align: center; padding-top: 60px;
         background: white; color: black; transition: 0.3s; }
  button { padding: 10px 18px; margin: 5px; cursor: pointer; border: 1px solid #555; }
  .card { width: 350px; margin: 30px auto; padding: 20px; border: 1px solid gray; }
&lt;/style&gt;
&lt;/head&gt;
&lt;body&gt;
  &lt;h1&gt;Choose a Theme&lt;/h1&gt;
  &lt;button onclick="setTheme('light')"&gt;Light&lt;/button&gt;
  &lt;button onclick="setTheme('dark')"&gt;Dark&lt;/button&gt;
  &lt;button onclick="setTheme('blue')"&gt;Blue&lt;/button&gt;
  &lt;button onclick="setTheme('green')"&gt;Green&lt;/button&gt;

  &lt;div class="card"&gt;
    &lt;h3&gt;Sample Card&lt;/h3&gt;
    &lt;p&gt;Click on any button above to change the color theme of this page.&lt;/p&gt;
  &lt;/div&gt;

&lt;script&gt;
function setTheme(theme) {
  var body = document.body;

  if (theme === "light") {
    body.style.background = "white";
    body.style.color = "black";
  } else if (theme === "dark") {
    body.style.background = "#222";
    body.style.color = "white";
  } else if (theme === "blue") {
    body.style.background = "#cce5ff";
    body.style.color = "#003366";
  } else if (theme === "green") {
    body.style.background = "#d4f5d4";
    body.style.color = "#1d5a1d";
  }
}
&lt;/script&gt;
&lt;/body&gt;
&lt;/html&gt;
```

Output:



4. Create Digital Clock using JavaScript

Code (clock.html):

```
&lt;!DOCTYPE html&gt;
&lt;html&gt;
&lt;head&gt;
&lt;meta charset="UTF-8"&gt;
&lt;title&gt;Digital Clock&lt;/title&gt;
&lt;style&gt;
  body { background: black; text-align: center; padding-top: 150px; }
  #clock { color: #00ff66; font-family: "Courier New", monospace; font-size: 70px; }
  #date { color: white; font-family: Arial, sans-serif; font-size: 20px; margin-top: 10px; }
&lt;/style&gt;
&lt;/head&gt;
&lt;body&gt;
  &lt;div id="clock"&gt;&lt;/div&gt;
  &lt;div id="date"&gt;&lt;/div&gt;

&lt;script&gt;
function showTime() {
  var now = new Date();
  var h = now.getHours();
  var m = now.getMinutes();
  var s = now.getSeconds();
  var ampm = h &gt;= 12 ? "PM" : "AM";

  h = h % 12;
  if (h === 0) h = 12;

  // add 0 in front if number is less than 10
  if (h &lt; 10) h = "0" + h;
  if (m &lt; 10) m = "0" + m;
  if (s &lt; 10) s = "0" + s;

  document.getElementById("clock").innerHTML = h + ":" + m + ":" + s + " " + ampm;
  document.getElementById("date").innerHTML = now.toDateString();
}

showTime();
setInterval(showTime, 1000);
&lt;/script&gt;
&lt;/body&gt;
&lt;/html&gt;
```

Output:



5. Create a todo list app in which user can add, delete and mark tasks as completed.

Code (todo.html):

```
&lt;!DOCTYPE html&gt;
&lt;html&gt;
&lt;head&gt;
&lt;meta charset="UTF-8"&gt;
&lt;title&gt;Todo List&lt;/title&gt;
&lt;style&gt;
  body { font-family: Arial, sans-serif; background: #f2f2f2; }
  .box { background: white; width: 380px; margin: 50px auto; padding: 25px; border: 1px solid #999; }
  input[type="text"] { width: 65%; padding: 8px; }
  button { padding: 8px 12px; cursor: pointer; }
  ul { list-style: none; padding: 0; }
  li { display: flex; align-items: center; justify-content: space-between;
       padding: 8px; border-bottom: 1px solid #ddd; }
  li span { flex: 1; margin-left: 8px; cursor: pointer; }
  li.done span { text-decoration: line-through; color: gray; }
&lt;/style&gt;
&lt;/head&gt;
&lt;body&gt;
  &lt;div class="box"&gt;
    &lt;h2&gt;Todo List&lt;/h2&gt;
    &lt;input type="text" id="taskInput" placeholder="Enter a task"&gt;
    &lt;button onclick="addTask()"&gt;Add&lt;/button&gt;
    &lt;ul id="taskList"&gt;&lt;/ul&gt;
  &lt;/div&gt;

&lt;script&gt;
function addTask() {
  var input = document.getElementById("taskInput");
  var text = input.value.trim();

  if (text === "") {
    alert("Please enter a task");
    return;
  }

  var li = document.createElement("li");

  var checkbox = document.createElement("input");
  checkbox.type = "checkbox";
  checkbox.onchange = function () {
    li.classList.toggle("done");
  };

  var span = document.createElement("span");
  span.innerText = text;

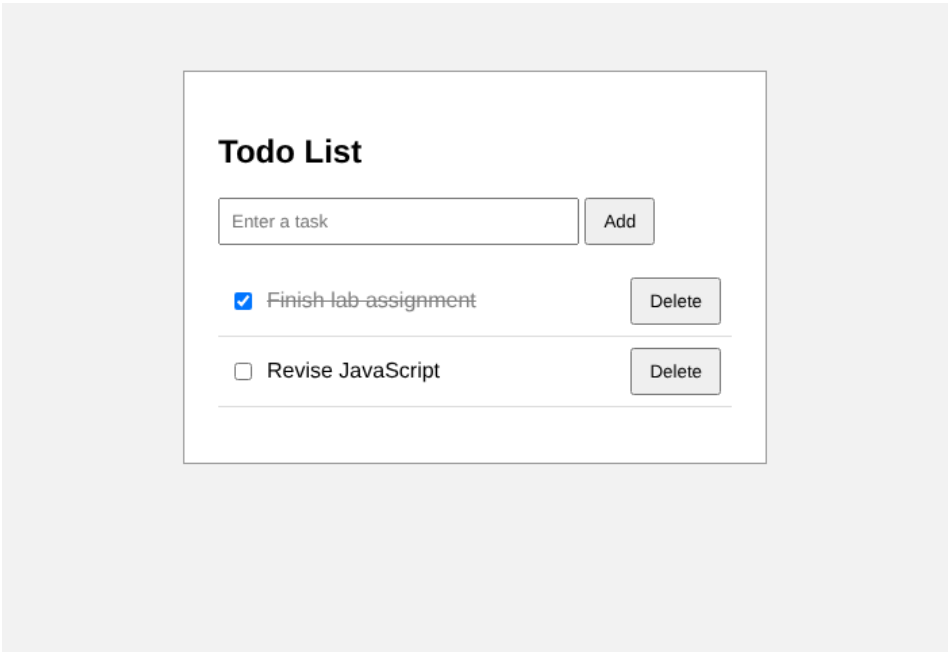
  var delBtn = document.createElement("button");
  delBtn.innerText = "Delete";
  delBtn.onclick = function () {
    li.remove();
  };

  li.appendChild(checkbox);
  li.appendChild(span);
  li.appendChild(delBtn);
  document.getElementById("taskList").appendChild(li);

  input.value = "";
}

// pressing Enter also adds the task
document.getElementById("taskInput").addEventListener("keydown", function (e) {
  if (e.key === "Enter") addTask();
});
&lt;/script&gt;
&lt;/body&gt;
&lt;/html&gt;
```

Output:


Todo List
Enter a task
Add
Finish lab assignment
Delete
Revise JavaScript
Delete